

Quantitative methods to assess sustainability (polytechnical foundation)

Where should I go?

Go to your first priority of case study.

If there is no seats left in the room of your first priority, go to your second priority and so forth.

			Entrance	
54				
Aquaculture Silvia Antunes				
		60B		61
		Milk Alessandro Parente		Anesthesia Mathilde Lei
				62
				Antifouling coating Mathilde Lei
				65
				Internet Simon Weiser
				66
				Microfabrication Simon Weiser
				69
				Decoupling Freshwater Ewa Palczynska
46		60A		
Marine tourism Marta Correia		Decarbonizing electricity Alessandro Parente		
45		43		42
Marine tourism Marta Correia		Materials for Davide Maccarone		Satellites Natacha Théron

Main course responsible



David Lusseau
Professor
DTU Aqua

Course co-responsibles



Elena Corella Puertas
Assistant professor
DTU Sustain



Stig Irving Olsen
Associate Professor
DTU Sustain

Coordinator



Niklas Braarud Jordal
Academic employee
DTU Sustain

Program for teaching session

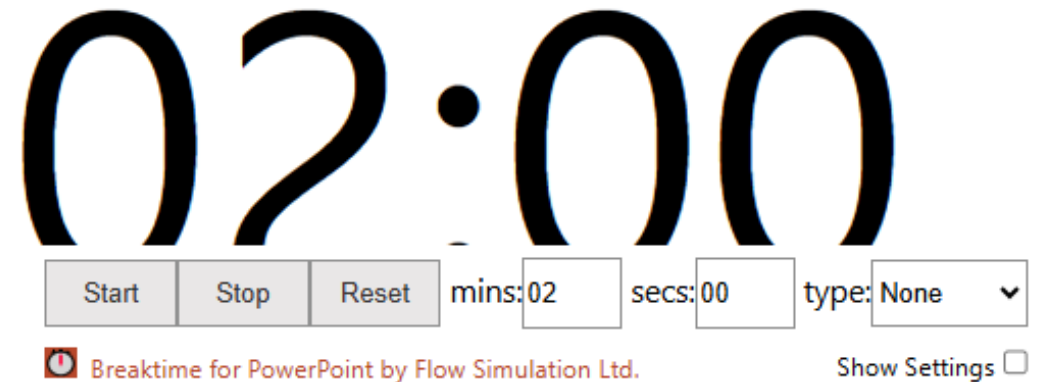
- Introduction (15 min)
 - Introduction of course responsables and coordinator
 - Introduction of teaching assistants
- Peer-to-peer discussions; *Problems, relevance and obligations* (20 min)
- Course formalities (35 min)
- Introduction to group formation (5 min)
- Group formation (25 min)
- (Did you not find a group? Let's meet in room 70)

Teaching assistants

- Supervising and guiding through both general modules and case study teaching
- Present during the 2 hour teaching sessions

Presentation of teaching assistants (2 min)

- Name
- Educational background
- Case study allocation(s)
- Prior experience with the course and quantitative sustainability assessment



1/4 Peer-to-peer discussion

How does the work of engineers influence environmental, social, and economic sustainability?

03:00

Start Stop Reset mins:3 secs:0 type: None ▾



Breaktime for PowerPoint by Flow Simulation Ltd.

Show Settings ☐

2/4 Peer-to-peer discussion

Why use quantitative methods to assess sustainability as an engineer?

03:00

mins: secs: type:

 Breaktime for PowerPoint by Flow Simulation Ltd.

Show Settings ☐

3/4 Peer-to-peer discussion

Question 1 from case study teaching assistant

- 1) Teaching assistant writes reflection question on the blackboard
- 2) Teaching assistant presents the relevance of the question to the case study specific subject
- 3) Students discuss the question in groups

04:00

Start Stop Reset mins:4 secs:0 type: None ▼

 Breaktime for PowerPoint by Flow Simulation Ltd.

Show Settings ☐

4/4 Peer-to-peer discussion

Question 2 from case study teaching assistant

Teaching assistant writes reflection question on the blackboard (1 min)

Teaching assistant presents the relevance of the question to the case study specific subject (1 min)

Students discuss the question in groups (3 min)

04:00

Start Stop Reset mins:4 secs:0 type: None ▼

 Breaktime for PowerPoint by Flow Simulation Ltd.

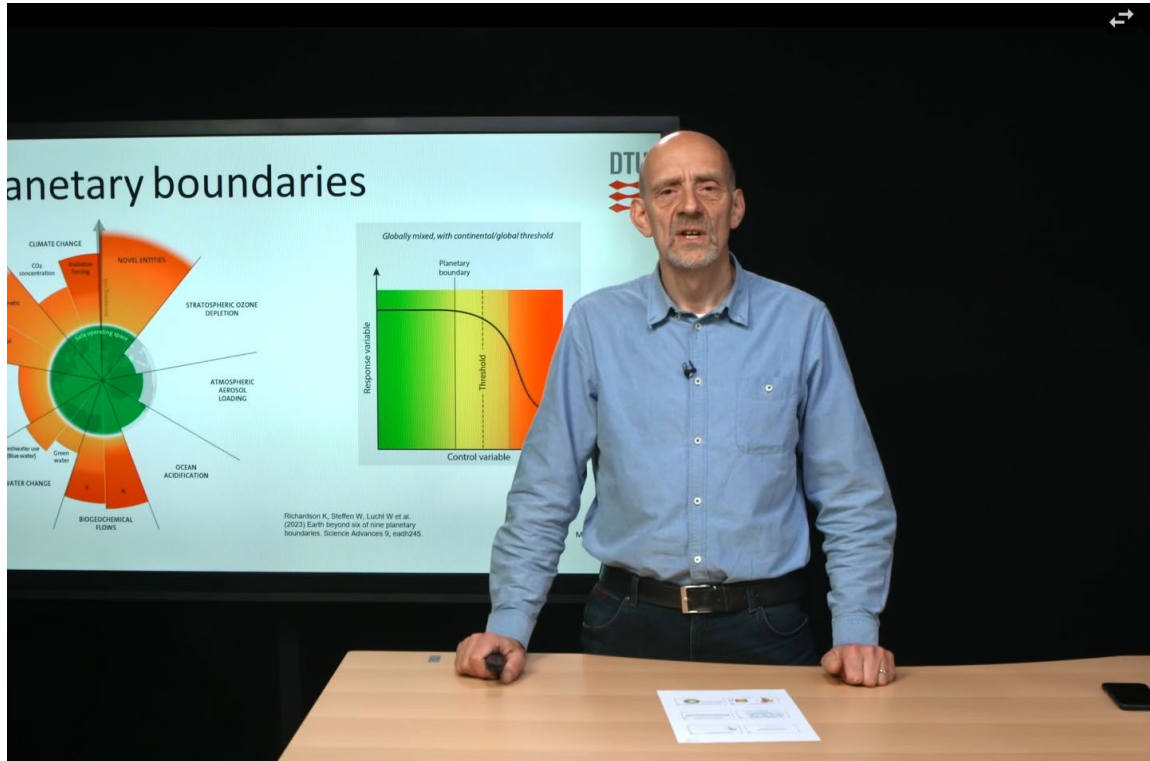
Show Settings ☐

Course schedule

- See *About the course* → *Schedule*

	Week	Subject	Teacher	Studies before	Teaching sessions
Part I - Problem definition and assessment framework	36 (1) 4/9	Problems, relevance and obligations	David Lusseau Elena Corella Puertas	See the Module description	See the Module description
	37 (2) 11/9	The assessment framework and functional units	Stig Irving Olsen	See the Module description	See the Module description
Case study	38 (3) 18/9	Problem statement and definition	Case study teachers	Case study dependent	Case study teaching
Part II Assessment methods	39 (4) 25/9	Carbon footprint and environmental indicators	Olivier Jean Joliet	See the Module description	See the Module description
	40 (5) 2/10	Health dimensions	Olivier Jean Joliet	See the Module description	See the Module description
	41 (6) 9/10	Economic dimensions	Georgios Manthos	See the Module description	See the Module description
	42				
	43 (7) 23/10	Planetary boundaries and absolute sustainability	Michael Zwicky Hauschild	See the Module description	See the Module description
	44 (8) 30/10	Resource management, scarcity and criticality	David Lusseau	See the Module description	See the Module description
Case study	45 (9) 6/11	Applying tools	Case study teacher	Case study dependent	Case study teaching
Part III Strategies	46 (10) 13/11	Energy systems and transition /decarbonization	Asger Bech Abrahamsen	See the Module description	See the Module description
	47 (11) 20/11	Circular economy	Tim C. McAlone Giovana Monteiro Gomes	See the Module description	See the Module description
	48 (12) 27/11	Sustainability transitions	David Lusseau	See the Module description	See the Module description
Case study	49 (13) 4/12	Sustainability assessment	Case study teacher	Case study dependent	Case study teaching

Weekly structure of general modules



- *Module descriptions:*

- *Studies before*

- Watch video lectures
- Reading literature
- Studying self-assessment (if any)

- *Teaching session*

- Exercises on topic
- Supervision and guidance from teacher(s) and teaching assistants

Teaching sessions

- 2 hours together with teacher and teaching assistants

General module teaching session:

- Focus on exercises:
 - Introduction to exercise and tool explanation
 - Follow up on exercises and interpretation
 - Guidance and supervision with teaching assistants
 - (Time to study case study)

Case study teaching session:

- Case study specific

Teaching rooms

- Same for general modules and case study teaching
- One teaching assistant allocated to each teaching room
- See DTU Learn, *ABOUT THE COURSE* → *Teaching rooms*

Exercises and data sheets

Carbon footprint tool

Functional Unit:
1 cup (=1 dl) of coffee

Transformation multiplier	1000	t/kg, kg/g...
Conversion factor	3,6	MJ/kWh
Conversion efficiency - Electricity	25%	
Conversion efficiency -Heat	80%	

System 1: Robusta coffee, DK electricity

Reference flow		Unit
Number of days per year	365	Days
N° of coffee drinkers per household	2	Persons
N° of coffee per person per day (#cup/pers-d)	2	Cups
Lifespan of Coffee maker (Yr)	10	Years
Coffee maker lifespan (#cup over lifetime)	14600	Cups
Weight of coffee maker	3	kg

Material	Energy content	Unit
Paper incineration mixture	16	MJ/kg
Plastic incineration mixture	42,5	MJ/kg

List of Processes	Process ID	Corresponding Eco-invent process	Location	Quantity per unit	Unit	Quantity per FU	Unit	Carbon footprint [kg CO2eq/Unit]	Carbon footprint [kg CO2eq/FU]	fraction carbon footprint	Humal health [min. lost/FU]	Human health [min. lost/FU]	fraction human health
Manufacturing (Materials + Processing)		Manufacturing (Materials + Processing)		per unit		per FU							
Coffee green beans production robusta (11.5g)	3069	coffee green bean production, robusta	RoW	0,0115	kg/serving	1,15E-02	kg	5,54	6,37E-02	62,07%	12,51	1,44E-01	-0,7%
Heat from natural gas for coffee roasting	10063	heat production, natural gas, at boiler condensing modulation	Europe without Switzerland	2,63	MJ/kg coffee	3,02E-02	MJ	0,07	2,19E-03	2,14%	0,00	1,23E-04	0,0%
Electricity for coffee roasting	7535	market for electricity, low voltage	DK	0,11	kWh/kg coffee	1,27E-03	kWh	0,19	2,34E-04	0,23%	0,10	1,28E-04	0,0%
Coffee machine (1 unit of 3 kg)	3057	market for coffee maker	GLO	1	unit	6,85E-05	unit	41,08	2,81E-03	2,74%	28,35	1,94E-03	0,0%
Paper Cup (8g paper)	12216	market for kraft paper	RER	0,008	kg/serving	8,00E-03	kg	0,55	4,38E-03	4,27%	0,21	1,68E-03	0,0%
Packaging (capsule 2g alu - aluminium)	523	market for aluminium, primary, ingot	EA Area, EU27 & EFTA	0,002	kg/serving	2,00E-03	kg	10,18	2,04E-02	19,85%	5,65	1,13E-02	-0,1%
Transportation		Transportation		km distance		per FU							
Coffee, transport, freight, lorry 16-32 metric ton (3500 km)	22338	transport, freight, lorry, 16-32 metric ton, diesel, EURO 4	RoW	3500	km	4,03E-02	metric ton*km	0,20	8,09E-03	7,89%	0,07	2,82E-03	0,0%
Coffee transport, freight, sea, container ship, (4500 km)	22446	transport, freight, sea, container ship, heavy fuel oil	GLO	4500	km	5,18E-02	metric ton*km	0,01	5,31E-04	0,52%	0,01	4,76E-04	0,0%
Use phase		Use phase		per cup		per FU							
Tap water (0.1 L/cup)	21695	market for tap water	Europe without Switzerland	0,1	kg/serving	1,00E-01	kg	0,00	3,02E-05	0,03%	0,00	2,04E-05	0,0%
Electricity hot water (0.12 kWh/L)	7535	market for electricity, low voltage	DK	0,12	kWh/liter	1,20E-02	kWh	0,19	2,22E-03	2,17%	0,10	1,19E-03	0,0%
Stand-by electricity (0.028 kWh/cup)	7535	market for electricity, low voltage	DK	0,028	kWh/serving	2,80E-02	kWh	0,19	5,19E-03	5,06%	0,10	2,78E-03	0,0%
Health impact or benefit of coffee consumption	100001	Health benefit		cup of coffee		1,00E+00	cup of coffee	0,00	0,00E+00	0,00%	-20,80	-2,08E+01	100,8%
Disposal		Disposal		kg/unit coffee maker		per FU							
Aluminium sent to sanitary landfill	23363	treatment of waste aluminium, sanitary landfill	CH	0,002	kg/serving	2,00E-03	kg	0,02	3,43E-05	0,03%	0,02	3,46E-05	0,0%
Coffee maker as plastic mixture incineration	23957	market for waste plastic, mixture	DK	3	kg/coffee maker	2,05E-04	kg	2,34	4,81E-04	0,47%	0,15	3,04E-05	0,0%
Paper cup sent to incineration	23743	market for waste packaging paper	DK	0,008	kg/serving	8,00E-03	kg	0,01	5,01E-05	0,05%	0,00	1,94E-05	0,0%
Avoided heat paper and plastic incineration	10063	heat production, natural gas, at boiler condensing modulation	Europe without Switzerland			-8,20E-02	MJ	0,07	-5,95E-03	-5,80%	0,00	-3,32E-04	0,0%
Avoided electricity paper and plastic incineration	7535	market for electricity, low voltage	DK			-9,50E-03	kWh	0,19	-1,78E-03	-1,72%	0,10	-9,43E-04	0,0%
								Total CO2eq/FU	0,102545		Total min. lost/FU	-20,634873	

System 2: Robusta coffee, Hard coal electricity

Reference flow		Unit
Number of days per year	365	Days
N° of coffee drinkers per household	2	Persons
N° of coffee per person per day (#cup/pers-d)	2	Cups
Lifespan of Coffee maker (Yr)	10	Years
Coffee maker lifespan (#cup over lifetime)	14600	Cups
Weight of coffee maker	3	kg

Material	Energy content	Unit
Paper incineration mixture	16	MJ/kg
Plastic incineration mixture	42,5	MJ/kg

List of Processes	Process ID	Corresponding Eco-invent process	Location	Quantity per unit	Unit	Quantity per FU	Unit	Carbon footprint [kg CO2eq/Unit]	Carbon footprint [kg CO2eq/FU]	fraction carbon footprint	Humal health [min. lost/FU]	Human health [min. lost/FU]	fraction human health
Manufacturing (Materials + Processing)		Manufacturing (Materials + Processing)		per unit		per FU							
Coffee green beans production robusta (11.5g)	3069	coffee green bean production, robusta	RoW	0,0115	kg/serving	1,15E-02	kg	5,54	6,37E-02	48,58%	12,51	1,44E-01	-0,7%
Heat from natural gas for coffee roasting	10063	heat production, natural gas, at boiler condensing modulation	Europe without Switzerland	2,63	MJ/kg coffee	3,02E-02	MJ	0,07	2,19E-03	1,67%	0,00	1,23E-04	0,0%
Electricity for coffee roasting	4422	electricity production, hard coal	DE	0,11	kWh/kg coffee	1,27E-03	kWh	1,08	1,37E-03	1,04%	0,14	1,71E-04	0,0%
Coffee machine (1 unit of 3 kg)	3057	market for coffee maker	GLO	1	unit	6,85E-05	unit	41,08	2,81E-03	2,15%	28,35	1,94E-03	0,0%

Assessment framework and functional units



Stig Irving Olsen
Associate professor
DTU Sustain

Carbon footprint and environmental indicators



Olivier Jolliet
Professor
DTU Sustain

Health dimensions



Olivier Jolliet
Professor
DTU Sustain

Economic dimension



Georgios Manthos
Postdoc
DTU Sustain

Planetary boundaries and absolute sustainability



Michael Hauschild
Professor
DTU Sustain

Ressource management, criticality and scarcity



David Lusseau
Professor
DTU Aqua

Energy systems and transition/decarbonization



Asger Bech Abrahamsen
Senior researcher
DTU Wind

Circular Economy



Tim McAloone
Professor
DTU Construct



Giovana Monteiro Gomas
Researcher
DTU Construct

Sustainability transitions



David Lusseau
Professor
DTU Aqua

Why the **STRUCTURE** of this course?

- *This is a cognitively demanding course:*
 - You have the unique opportunity to be exposed to the diverse quantitative challenges faced across disciplines and sectors represented in Departments at DTU
 - It is your role to create a ‘red thread’ through this material for challenges that are relevant to you through the case study work
 - Do not anticipate to have to use all teaching modules for all case studies
 - We believe that student-led innovation is critical to meet the sustainability challenges we have

Why this course CONTENT?

Providing a sustainability foundation for all engineers.

Why DIFFERENT TEACHERS in each module?

Insight into different sustainability perspectives and expertise at DTU.

Why QUANTITATIVE methods?

You cannot manage what you cannot measure.

Why for ALL ENGINEERS?

Engineers have a special role to play in sustainability practices.

Generative AI when studying

DTU embraces the use of artificial intelligence (generative AI) and focuses on its potential to enhance learning and well-being. DTU allows the use of generative AI in teaching.

Course specific:

- Always explicitly declare if any part of your studies is not your original work. Failure to do so may be considered cheating.
- Use of AI in peer feedback is not allowed.
- Use AI as an assistant and focus on developing your critical thinking skills.

Course description

kurser.dtu.dk/

Supplementary registration period closes Wednesday 23:59 in the second week of the semester.
All correspondence concerning the course must be sent to course12106@dtu.dk

Course information	
Danish title	Kvantitative metoder til at vurdere bæredygtighed (Polyteknisk grundlag)
Language of instruction	English
Point(ECTS)	5
Course type	MSc Polytechnical foundation (MSc), see more General competence course, MSc. Eng., Bioinformatics
Schedule	Autumn E3B (Fri 13-17) The focus in the course is on student-centered teaching and active learning. For the course's teaching approach, this means that students watch the video lectures and read articles prior to the teaching session, and the session itself is concentrated on students' active interaction with teachers and exercises. Each teaching session during the 13 week period is scheduled for 2 hours.
Location	Campus Lyngby The teaching sessions will take place at Lyngby Campus. For a smaller subset of students, it may be that the teaching will take place at the location where the respective candidate is based (e.g., Hirtshals and Kalundborg).
Scope and form	The student is expected to study video lectures and read articles before the in-class teaching. In order to create time for active learning with exercises together with the teachers. The in-class teaching is therefore focused on studying exercises and doing calculations on sustainability. Supervision and feedback in the in-class teaching will be with teaching assistants (13 out of 13 weeks), module teacher (10 out of 13 weeks) and case study teachers (3 out of 13 weeks).
Duration of Course	13 weeks
Date of examination	The exam is held on a special course-specific exam day (search for course number in DTU's overview of exam dates). Please note that the exam placement is scheduled for the last part of the examination period. The re-exam is scheduled in accordance with the ordinary exam for the course.

General course objectives
This course provides the understanding and quantitative tools needed by engineers and scientists to create engineering solutions for a sustainable society. The students learn about terms, definition, and units related to sustainability, the environmental, social and economic dimensions of sustainability and the importance of geographical and temporal scale when considering engineering solutions. Focus will be placed on understanding sustainability and developing sustainable solutions from an industrial perspective.
Learning objectives
A student who has met the objectives of the course will be able to: - Describe and critically reflect on the environmental, social, and economic considerations and challenges to creating 'sustainable engineering solutions' in practice across multiple geographical and temporal scales. - Critically evaluate sustainability in current practice across the three Sustainability Pillars; environment, society and economy. - Analyze and quantify, from a life cycle perspective, the sustainability of a chosen product or service through a group case study using the sustainability tools introduced during the course. - Apply current state-of-the-art tools to quantify sustainability whilst considering the complex interlinkage of sustainability challenges currently faced within climate change and the sources impacting climate change, effects on human health and the environment, effects on biodiversity of ecosystems, and the scarcity of certain resources. - Explain the uncertainties associated with sustainability assessments. - Explain the importance of a systemic perspective when assessing the sustainability of all technology and practices considering the current and future technological, environmental, social, and economic landscape. - Describe main components of an absolute environmental sustainability assessment. - Critically engage with the principles of life cycle assessment (LCA) in practice by evaluating the outputs of several life cycle assessments. - Describe the principle of Circular Economy and how that affects societal flows of resources.
Content
Terms, definitions and units describing sustainability, impact challenges to sustainable engineering solutions (Emission and climate change, Impacts on human, environment, scarcity of resources), social and economic dimensions of sustainable development, absolute environmental sustainability assessment, circular economy (definitions, reasons, and drivers), principle of life cycle assessments, quantitative tools for evaluating sustainability.

A student who has met the objectives of the course will be able to:

- *Describe and critically reflect on the environmental, social, and economic considerations and challenges to creating 'sustainable engineering solutions' in practice across multiple geographical and temporal scales.*
- *Critically evaluate sustainability in current practice across the three Sustainability Pillars; environment, society and economy.*
- *Analyze and quantify, from a life cycle perspective, the sustainability of a chosen product or service through a group assignment using the sustainability tools introduced during the course.*
- *Apply current state-of-the-art tools to quantify sustainability whilst considering the complex interlinkage of sustainability challenges currently faced within climate change and the sources impacting climate change, effects on human health and the environment, effects on biodiversity of ecosystems, and the scarcity of certain resources.*
- *Explain the uncertainties associated with sustainability assessments*
- *Explain the importance of a systemic perspective when assessing the sustainability of all technology and practices considering the current and future technological, environmental, social, and economic landscape.*
- *Describe main components of an absolute environmental sustainability assessment.*
- *Critically engage with the principles of life cycle assessment (LCA) in practice by evaluating the outputs of several life cycle assessments.*
- *Describe the principle of Circular Economy and how that affects societal flows of resources.*

(Course description)

Exam dates

- Special course specific exam date (the exam schedule is not same as the teaching schedule of the course)
- See course description
- See overview of exams
 - <https://student.dtu.dk/en/exam/exam-dates>

Exam prerequisites

- Submit task 1 and 2
- Give peer feedback on task 1 and task 2

Formalities of task 1 SUBMISSION and PEER FEEDBACK

FORMALITIES OF SUBMISSION

Task 1 is submitted in *FeedbackFruits*-element under CASE STUDIES and *Task 1*.

- Must cover the Task 1 described in chosen *Case study description*.
- Must be submitted by only one student from each group.
- Must include a *Table of Contributions* with an overview of which group members contributed to which parts of the different tasks. The table of contributions must be agreed upon by all group member before attaching it to the submission.

Unless anything else is described in the case study description, the following is applicable:

- Should be a maximum of 7200 characters - corresponding to three normal pages (excluding bibliography and tables).
- All use of language models should be declared.
- No specific format is required.

See deadline for submission under *0.1 About the course* and *Deadlines*.

FORMALITIES OF PEER FEEDBACK

Task 1 peer feedback is given in *FeedbackFruits*-element under CASE STUDIES and *Task 1*.

- The peer feedback must be done by each student individually.
- Each student will receive a submission from two other groups within the same case study.
- Each student must give peer feedback to two other group submissions.
- The peer feedback is given based on the criteria described.
- All criteria must be addressed to complete the peer feedback.
- The feedback received is revised by the study group to consider potential changes of the task 1 submission (final version of task 1 is included as an annex in task 2 and task 3 submission to enable the peers and teacher to follow any changes that have been made as a result of the peer feedback).

Use of AI in peer feedback is not allowed.

See deadline for peer feedback under *0.1 About the course* and *Deadlines*.

Formalities of task 2 SUBMISSION and PEER FEEDBACK

FORMALITIES OF SUBMISSION:

Task 2 is submitted in *FeedbackFruits-element* under CASE STUDIES and Task 2.

- Must cover the Task 2 described in chosen case study description.
- Must be submitted by only one student from each group.
- Must include a *Table of Contributions* with an overview of which group members contributed to which parts of the different tasks. The table of contributions must be agreed upon by all group member before attaching it to the submission.
- Must include an annex with Task 1 (as a foundation for peers to understand Task 2 when giving peer feedback)

Unless anything else is described in the case study description, the following is applicable:

- Should be a maximum of 9600 characters - corresponding to four normal pages (excluding bibliography and tables).
- All use of language models should be declared
- No specific format is required.

See deadline for submission under ABOUT THE COURSE and *Deadlines*.

FORMALITIES OF PEER FEEDBACK

Task 2 is submitted in *FeedbackFruits-element* under CASE STUDIES and *Task 2*.

- The peer feedback must be done by each student individually.
- Each student will receive a submission from two other groups within the same case study.
- Each student must give peer feedback to two other group submissions.
- The peer feedback is given based on the criteria described in *FeedbackFruits*.
- *All* criteria must be addressed to complete the peer feedback.
- The feedback received is revised by the study group to consider potential changes of the task 2 submission (final version of task 2 is included as an annex in the task 3 submission to enable the teacher to follow any changes that have been made as a result of the peer feedback).

Use of AI in peer feedback is not allowed.

See deadline for peer feedback under ABOUT THE COURSE and *Deadlines*.

Formalities of task 3 submission

- Task 3 is submitted under CASE STUDIES and Task 3.
- Must cover the Task 3 described in chosen case study.
- Must include a *Table of Contributions* with an overview of which group members contributed to which parts of the different tasks. The table of contributions must be agreed upon by all group member before attaching it to the submission.
- Must include an annex with Task 1 and Task 2 (as a foundation for teachers to understand Task 3).
- Must be submitted in by only one student from each group.

Unless anything else is described in the case study description, the following is applicable:

- Should be a maximum of 9600 characters - corresponding to four normal pages (excluding bibliography and tables).
- All use of language models should be declared
- No specific format is required.

See deadline for submission under ABOUT THE COURSE and *Deadlines*.

Where can I read about the formalities?

CASE STUDIES →

FORMALITIES: SUBMISSION AND PEER FEEDBACK

- Use of generative AI
- Schedule
 - Completed
- ▼ CASE STUDIES
 - 7/182 Completed
 - ▼ FORMALITIES: SUBMISSION & PEER...
 - 5/9 Completed
 - ▶ Task 1
 - 2/4 Completed
 - ▶ Task 2
 - 2/4 Completed
 - ▶ Task 3
 - 1/1 Completed
 - ▶ Anesthesia
 - 0/20 Completed
 - ▶ Antifouling coating
 - 0/6 Completed
 - ▶ Decoupling Freshwater

Deadlines

- See *ABOUT THE COURSE* → *Deadlines*

Special examination conditions

Ordinary written exams – which are arranged by the central study administration

- If you are granted special exam conditions, They are registered automatically to ordinary written exams
- The student does not need to contact the lecturer – just check the exam-plan. If any mistakes, please contact the study administration, studadm@adm.dtu.dk.

Case study	Submission	Peer feedback
Task 1	Submission in group Complete/Not complete	Individual peer feedback on two submissions on defined criteria Complete/Incomplete
Task 2	Submission in group Completed/Not completed	Individual peer feedback on two submissions on defined criteria Complete/Incomplete
Task 3	Submission in group Counts 40% in grading	No

General modules	
Multiple choice exam	Focus on general modules Conceptual questions and calculation-based questions You must have at least 50% correct in order to pass the course Counts 60 % in grading

Assessment - Grades

DK 7-point grading scale	EU ECTS grading scale	Meaning
12	A	Pass
10	B	Pass
7	C	Pass
4	D	Pass
02	E	Pass
00	Fx	Failed
-3	F	Failed
IG		Exam requirements not fulfilled

Contact

General questions

Please contact course responsables and coordinator by sending an e-mail to your relevant course e-mail address:

course12106@dtu.dk

Questions about the case studies

Please contact the case study teacher by e-mail.

Questions about the general modules

Please contact the relevant module teacher.

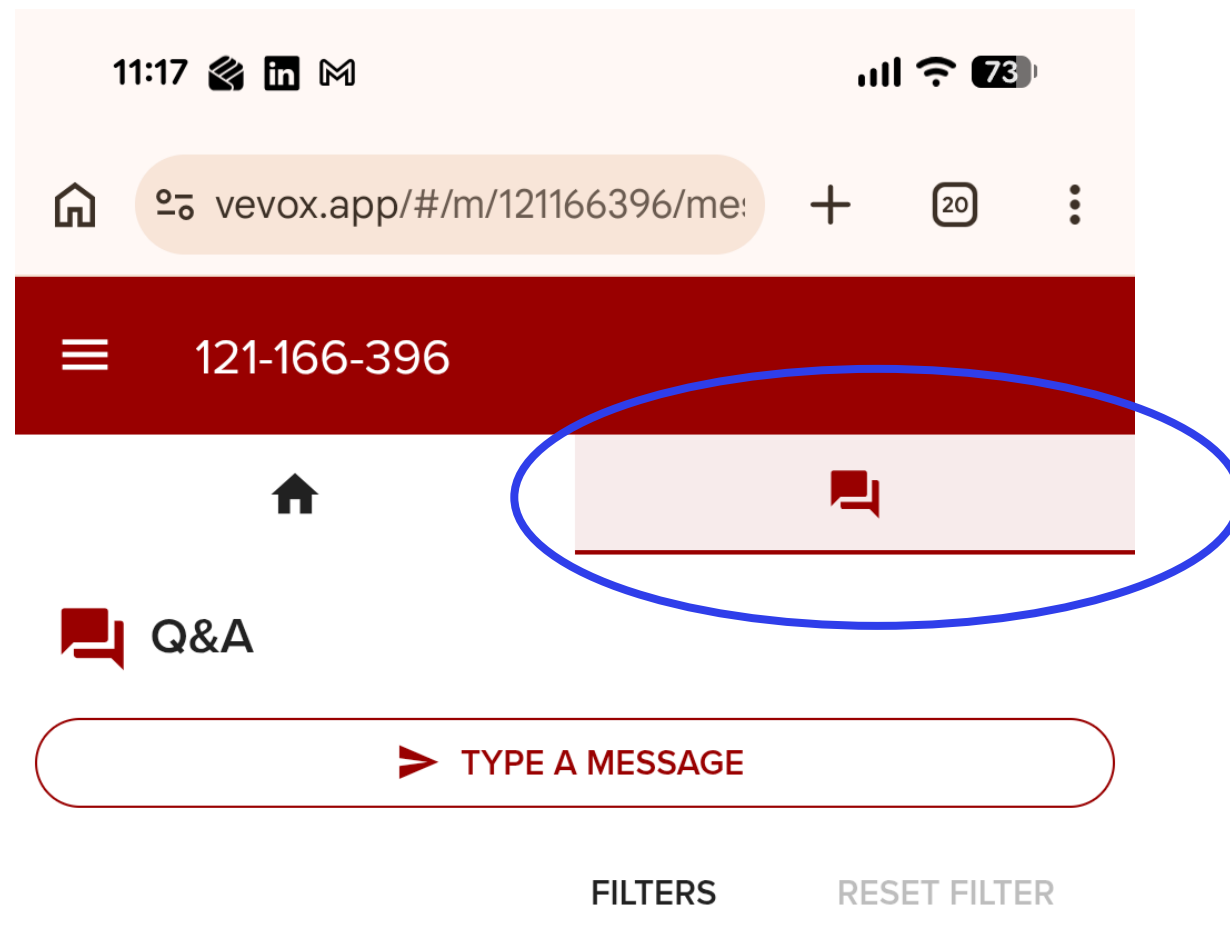
Join the Vevox session

Go to **vevox.app**

Enter the session ID: **121-166-396**

Or scan the QR code





Description of the process of group formation

Step 1: Form a group of 5 peers

- 1.1: Contact and talk with peers in different case study rooms
- 1.2: Or use the discussion forum "*Find a group*" on DTU Learn

Step 2: Discuss

- 2.1: Discuss the *Group formation questions* (next slide)

Step 3: One student from the group completes the group formation survey

- 3.1: Survey opens under *W1 Teaching session* 15 min before week 1 teaching session ends.
- 3.2: Survey closes 23:59 on the day of week 2 teaching session
- 3.3: The final case – and group allocation will be announced on DTU Learn before week 3

Allocation principle: Time of submission.

When you have found a group: Discuss group formation questions

Case study priorities

- Agree on 4 priorities
- Send in your priorities (on computer)
 - Link in *W1, teaching session* – opens 15 min before the teaching session ends

Planning of studying of task 1, 2 and 3

- When should we agree to meet to discuss the case study task 1, 2 and 3 submissions?
- When should we meet about the revision of task 1 and 2 after having received peer feedback? (Peer feedback is available one week after task submission)
- How we would we like to conduct the table of contribution for each task?

How would we like to collaborate on case study submissions?

- *Do we meet up to discuss and write simultaneously, or do we write specific sections independently and discuss later?*
- *How often do we meet?*
- *Do we meet online, on campus or other?*
- *Tools we use to collaborate?*

			Entrance	
54				
Aquaculture Silvia Antunes				
	60B		61	62
	Milk Alessandro Parente		Anesthesia Mathilde Lei	Antifouling coating Mathilde Lei
			65	66
			Internet Simon Weiser	Microfabrication Simon Weiser
46	60A		69	
Marine tourism Marta Correia	Decarbonizing electricity systems Alessandro Parente		Decoupling Freshwater Ewa Palczynska	
45		43	42	
Marine tourism Marta Correia		Materials for Davide Maccarone	Satellites Natacha Théron	

See you next week

Or in room 70 20:05, if you are still searching for a group

Did you not find a group? Let's meet in room 70

- Meet other students who have not found a group yet
- Ask *group formation*-questions to coordinator

